Ashrafu Ally

+1 (607) 389-5591 | ara12@alfred.edu | linkedin.com/in/ashrafu-ally | Alfred, NY

Education

Alfred University — Alfred, NY

B.S. Materials Science and Engineering; Minor in Computer Science — *Expected May 2029*

- **Distinctions:** Presidential Scholar; Alfred University Honors Program; Dean's Scholarship ($23,000); Trustee Grant ($31,000); Dean's List (3.83 GPA, Freshman Fall)
- **Relevant Coursework:** Introduction to Computer Science I, Data Structures and Algorithms, Calculus I–II, General Chemistry, Physics I–II

Technical Skills

Programming & Tools: Python, NumPy, Matplotlib, Data Structures & Algorithms, Git/GitHub, Jupyter

Computing Infrastructure: High-Performance Computing (HPC), Linux Command-Line Interface, Job Scheduling & Automation

Scientific Computing & Simulation: LAMMPS, OVITO, DFTpy, Molecular Dynamics Simulation

Materials & Mathematics: Materials Characterization, Glass & Amorphous Structures, Calculus, Linear Algebra, Differential Equations

Experience

Undergraduate Research Assistant | Summer Undergraduate Research Fellow (SURF) — Jan. 2026 – Present

Alfred University, Computational Materials Science Lab (Dr. Zachary Knepp) — *Alfred, NY*

- Conduct computational materials research simulating the manufacturing of metallic glasses toward ultra-high-strength, lightweight, corrosion-resistant structural materials.
- Scaled and executed molecular dynamics simulations on the **San Diego Expanse HPC cluster** via the Linux command line; used **LAMMPS** to model amorphous structures.
- Fabricated a stable multi-component **Al-Mg (67/33%) metallic glass** using continuous ultra-fast temperature ramping at a quenching rate of 10^{11} K/s.
- Deployed **DFTpy** scripts to initialize electronic structure environments and optimize simulation jobs; customized open-source MD scripts from GitHub and wrote custom **Python** code to automate data extraction.
- Evaluated phase transitions, atomic packing, and structural evolution during rapid solidification using **OVITO**; presented findings via a research poster presentation.

Resident Assistant — Aug. 2026 – Present

Alfred University Residential Life — *Alfred, NY*

- Serve as a campus leader and primary resource for undergraduate students, fostering a safe, inclusive residential community; balance administrative duties with mentorship supporting student retention.

Fellowships & Programs

Fellow – Applied AI Engineering Pathway — 2026 – Present

CodePath — *Remote*

- Selected for a competitive program building applied software engineering skills in AI-assisted development workflows, strengthening core CS fundamentals.

Fellow – Technical Interview Prep — 2026

CodePath — *Remote*

- Completed a structured curriculum on data structures, algorithms, and technical interview problem-solving.

Fellow — 2026 – Present

ColorStack — *Alfred University Chapter*

- Member of a national program supporting Black and Latinx students pursuing careers in computing, including technical mentorship, peer study sessions, and career development.

Leadership & Campus Involvement

Chairperson, Student Employment Committee — Aug. 2026 – Present

Student Senate, Alfred University — *Alfred, NY*

- Lead committee discussions with students and university stakeholders to improve communication and processes around student employment.

Secretary & Content Creator — Mar. 2026 – Present

Materials Advantage & Keramos — *Alfred, NY*

- Manage communication, event documentation, and outreach for materials science and ceramics-related programming.